

UNIT 4: Weather Analysis Products

e. Oceanography, Tropical, & Space Weather

The goal of this objective will be to show you have the capability to perform a basic analysis of oceanic, tropical, and space weather data. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly identify and label various weather features discussed throughout this objective. These foundational analysis concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

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Introduction to Oceanography

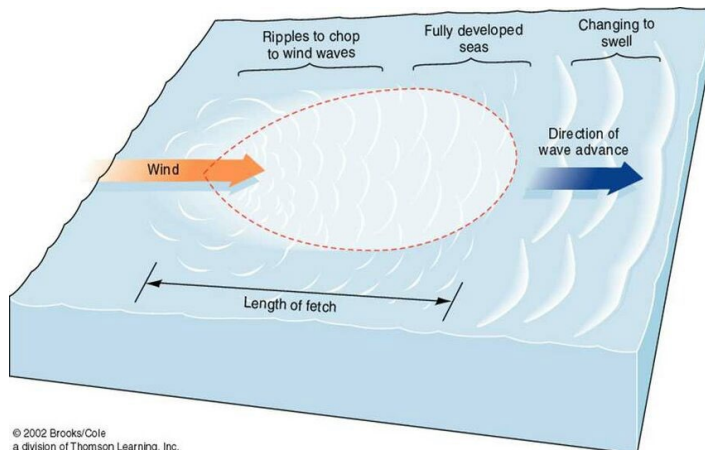
Oceanic Processes

Oceanography is important to military forecasters regardless of their branch of service. Oceanography will be discussed throughout this course beginning with the horizontal and vertical transportation of water within the Earth's oceans.

Waves

Fetch

A Fetch is the uninterrupted horizontal distance over which wind blows across a body of water in a constant direction, generating waves. The longer the fetch, the more energy the wind transfers to the water, resulting in larger, more powerful, and more organized waves with greater height and longer periods. Factors influencing fetch length include the size of the water body, surrounding landmasses, and the wind's consistent direction and consistent speed.



Wind Waves

Wind waves are ripples on the ocean's surface created by localized winds and are referred to as such while they remain within the region they were initially generated. The wave movement is the same as the wind direction and wind speed determines the wave's height. Generically

speaking, light winds produce small waves, while stronger winds produce higher waves. The time between consecutive waves is referred to as the “wave period” and is measured in seconds.



Swell Waves

Swell waves are long, low, and organized waves that form from energy transferred to the water by wind waves, often from storms far out at sea. Their size depends on the residual; uninterrupted energy created from the speed of winds associated with the storm.



Tsunami Waves

A tsunami is a series of extremely long waves caused by a large, sudden displacement of the ocean such as an undersea earthquake or volcanic eruption.



Exercise / Practice: Waves

Tides

Tides are cyclic rises and falls of seawater, influenced heavily by the moon and the sun.

Tide Types

The interaction between the Sun, Moon, and Earth as well as the Earth's land to water distribution creates different variations in water levels along coastal areas. These water levels are identified as a cycle of four transition periods:

High Tide: The period when water levels are at their highest and greatest expanse inland.

Ebb Tide: The period when high tide transitions to low tide as water begins to flow out to sea

Low Tide: The period when water levels are at their lowest and retreat the farthest offshore.

Flood tide: The period of when low tide transitions to high tide as water begins to flow towards land.

Moon's Influence

The moon determines the tide day (a.k.a. tide period) and tide height. The tide day is 24 hours and 50 minutes, which is equal to the time it takes the Moon to rotate around the Earth. During this time, oceans experience up to two high tides and two low tides. Tide height changes based on the Moon's position to the Earth and surface of the ocean.

High Tide: High tides occur when the Earth and Moon axis is parallel to the ocean. Oceans "facing" the moon have a strong gravitational attraction and higher high tides.

Low Tide: Oceans facing away from the moon have a weak gravitational attraction and a lower high tide.

Sun's influence

The Sun causes tide heights to increase and decrease. While its influence is minimal most of the time, there are two instances the Sun causes greater changes in the tide height. These are during spring and neap tides.

Spring Tide: Spring tides occur when the Sun and the Moon are parallel to Earth. This only occurs every 14 to 15 days during a new or full Moon. During spring tides, the range between tide heights increase creating high tides higher than average and low tides lower than average.

Neap Tide: Neap tides occur when the Sun and the Moon are perpendicular to the Earth. This also only occurs every 14 to 15 days during the first and last quarter Moon. During neap tides, the range between tide heights decrease creating high tides lower than average and low tides higher than average.

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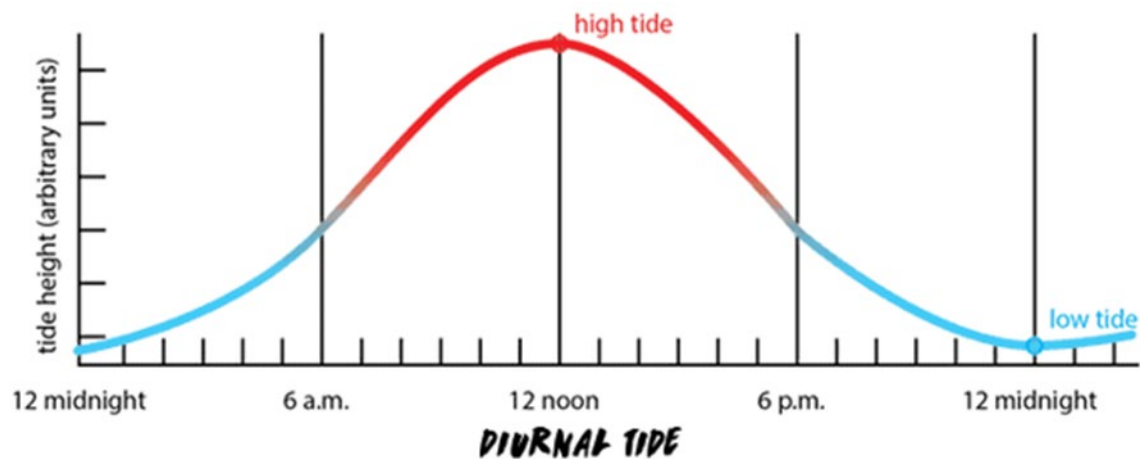
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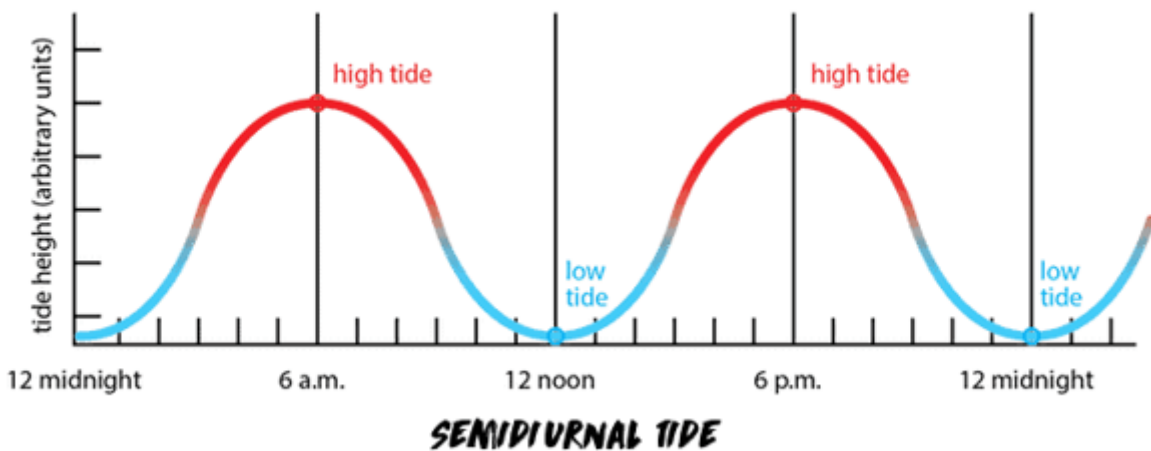
Flood tide: The period of when low tide transitions to high tide as water begins to flow towards land.

Diurnal Tide: It has one high and one low tide every lunar day. Example locations are the Gulf of America and Southeast Asia.



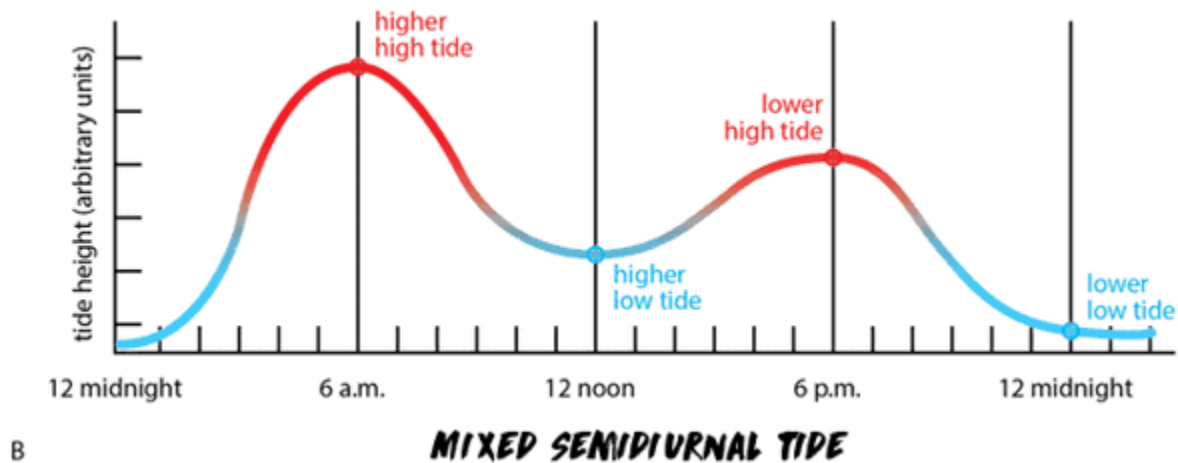
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Semi-diurnal Tide: It consists of two high and two low tides of close to equal height per tide day. They occur along the U.S. east coast and Europe.



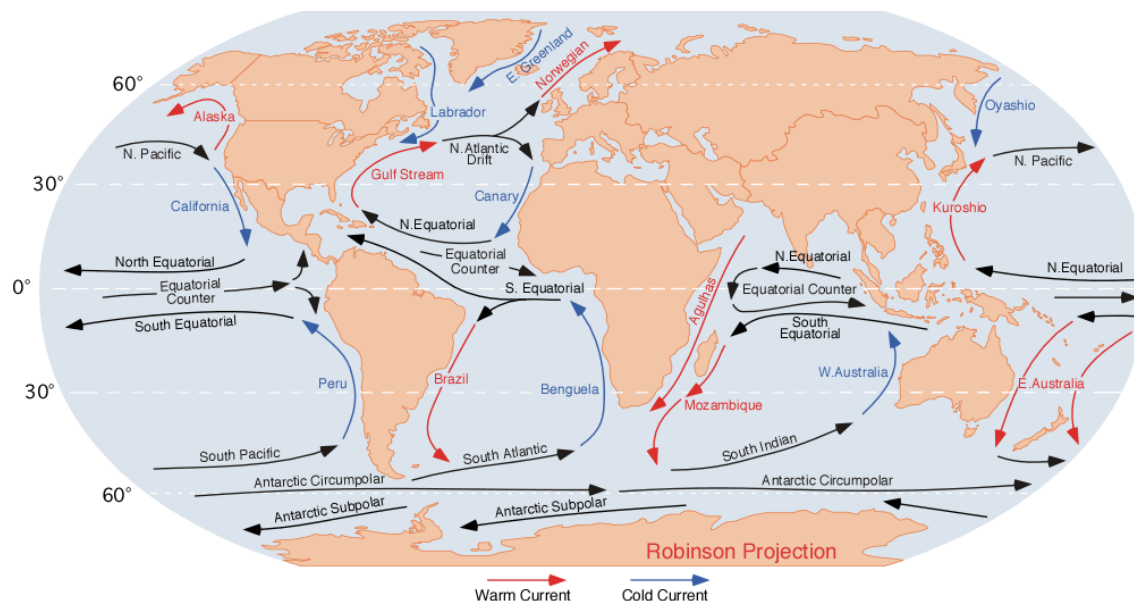
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Mixed Tide: It also consists of two high and two low tides per tide day, but they have different tide heights. One high tide is higher than the other high tide and one low tide is lower than the other low tide. The U.S. and Canadian west coasts have mixed tides. **Longshore current:** the longshore current is a current that runs parallel to the coast that can force landing craft to broach, an involuntary change in a vessel's course. To predict the direction of the longshore current requires forecasters to orientate their terrain to their waves. Imagine a North-South running beach and Waves coming from the Southwest moving Northeast. The Longshore current would be moving North Parallel to the beach as the waves energy is forced to continue in that direction. This can all be forecasted based on buoy data, manual observations, verified models, and satellite imagery.



Currents

Ocean currents are horizontal movements of seawater. Specifically, they are large features extending across the oceans hundreds of miles wide and thousands of miles long.



Current Intensity

A current's intensity has two components. The first component is its speed or drift, which is measured in knots. Conceptually, a fast drift is a strong current while a slow drift is a weak current. The second component is the current's direction of movement or set using compass points. For example, a current with a north set is moving towards 360 degrees while a current with a west set is moving towards 270 degrees.

Temperature

Cold currents occur where colder water moves into warmer water. This occurs off the west coasts of continents in the middle latitudes and off the east coasts of continents in the high latitudes. The Canary, Labrador, and East Greenland currents are examples cold currents in the Atlantic Ocean. Warm currents are off the east coast of continents in the middle latitudes and off the west coast of continents in the high latitudes. In the Atlantic Ocean, the Gulf Stream and Norwegian currents are warm currents.

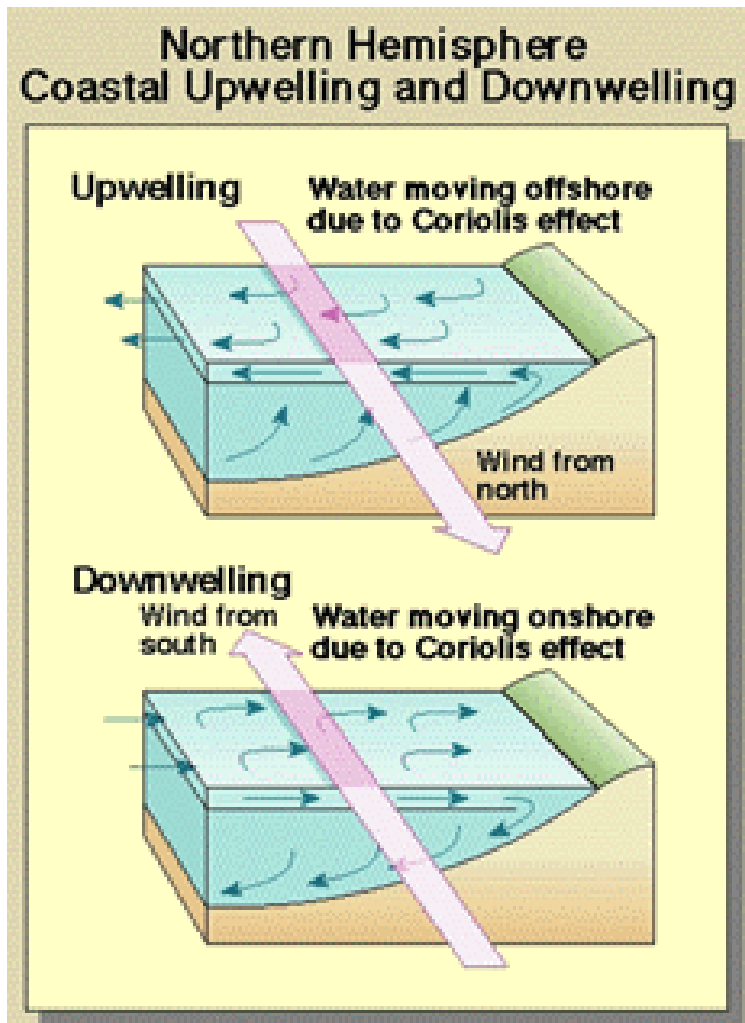
Subtropical Gyres

Subtropical gyres are large oval or circular ocean currents near 30 degrees latitude. They are permanent features that rotate anti-cyclonic in the Northern hemisphere and cyclonic in the Southern hemisphere. There are five subtropical gyres. The North Pacific and North Atlantic gyres in the northern hemisphere and the South Pacific, South Atlantic, and Indian Ocean gyres in the southern hemisphere.

Coastal upwelling and downwelling

Coastal Upwelling: Coastal upwelling is the forced upward movement of cold, sub-surface water to the ocean's surface. In the Northern Hemisphere, it occurs where northerly wind parallels the coastline. The water along the coast turns to the right due to Earth's rotation which moves the warm surface water away from the coast allowing colder subsurface water to rise and replace it.

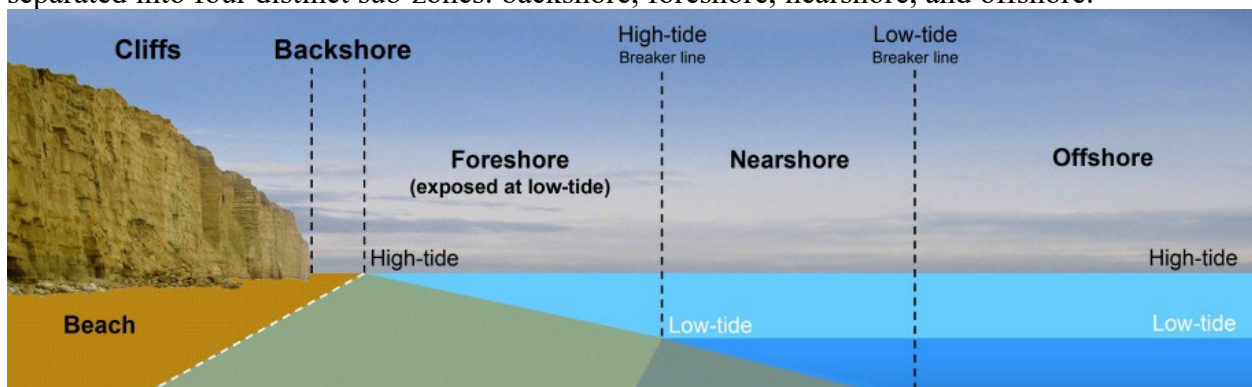
Coastal downwelling: Coastal downwelling occurs where warm, surface water is forced to sink. In the Northern Hemisphere, it occurs where southerly wind parallels the coastline. The water along the coast turns to the right due to Earth's rotation which moves the surface water toward the coast. This causes an accumulation of mass, and the weight of the water forces it to sink.



Exercise / Practice: Tides / Currents

Littoral Zones

The littoral zone describes the sea state extending from the shoreline out to the point where the ocean floor no longer has any impact to surface waters. No matter the size, the littoral zone is separated into four distinct sub-zones: backshore, foreshore, nearshore, and offshore.



Backshore

The backshore is considered dry, unless influenced by extreme weather features such as hurricanes or tsunamis. Located between the high tide mark and coastal vegetation that act as a protective barrier for oceanographic events that may affect the coastline. It is characterized by dry sand and pebbles.

Foreshore

The foreshore is the area of the beach ranging from low tide and high tide marks. This zone is constantly shifting as waves change the shape of the beach and is characterized by wet sand or mud.

Nearshore

The nearshore is the area from the low tide mark to where waves begin to break (tops of waves begin to white caps). The small area where waves begin to break is referred to as the surf zone. Within the surf zone lies the longshore current, a current that runs parallel to the coast that can force landing craft to broach, an involuntary change in a vessel's course. In the nearshore, there will be sandbars and breaker bars formed by the interaction of waves.

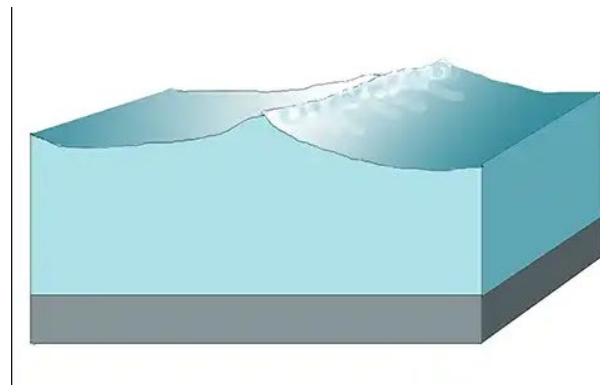
Offshore

The offshore describes the area of deeper water beyond the point where waves start breaking and have less interaction with the ocean floor, allowing for waves to maintain their regular shape and size.

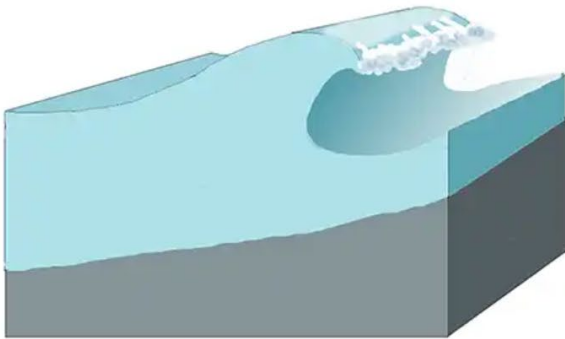
Breakers

As the swell waves move towards the nearshore environment, the interaction of the ocean floor and the energy generated by the waves cause it to become unstable and break. Breaker height and breaker type are the most important surf factors for determining the feasibility of an amphibious beach operation. The interaction of the swell waves, slope of the ocean floor, wave period, and surface winds will cause one of three types of breaker waves: Spilling, Plunging, and Surging. The three types of breakers are described in the following.

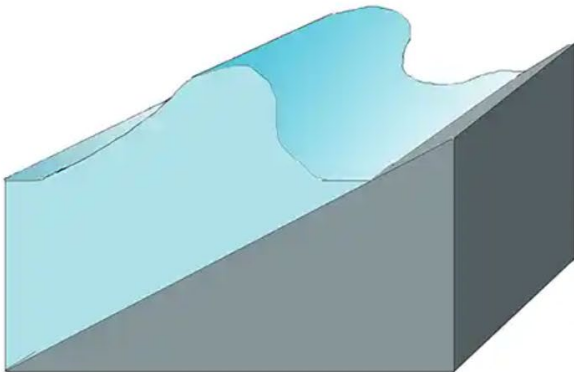
Spilling: Spilling breakers normally occur on nearly horizontal beaches. The crest slides down the face of the wave forming foam and gradually releasing energy over a wide area.



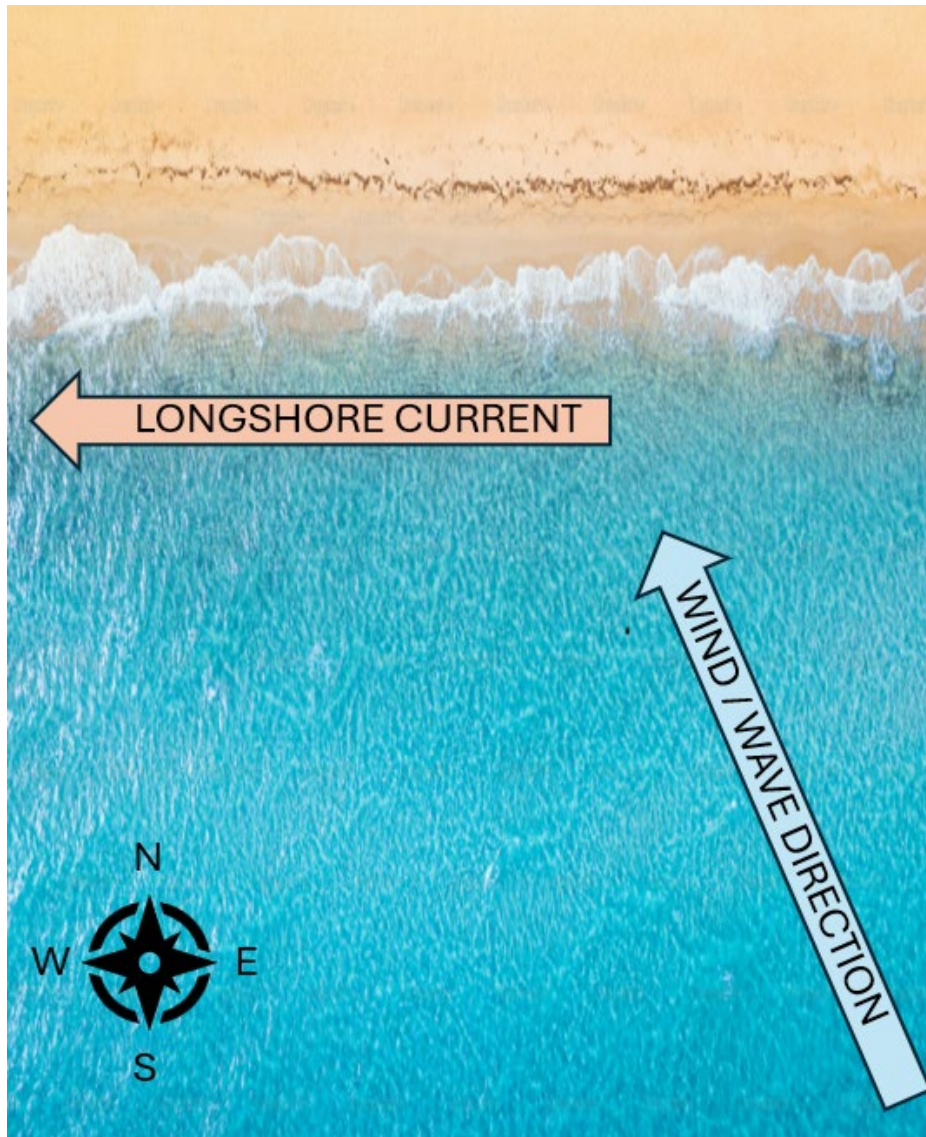
Plunging: The slope of the ocean floor is moderately steep, forcing the wave height to increase. This instability causes the top of the wave to crash directly to the bottom. The crest plunges over into the trough with a sudden release of energy over a narrow area.

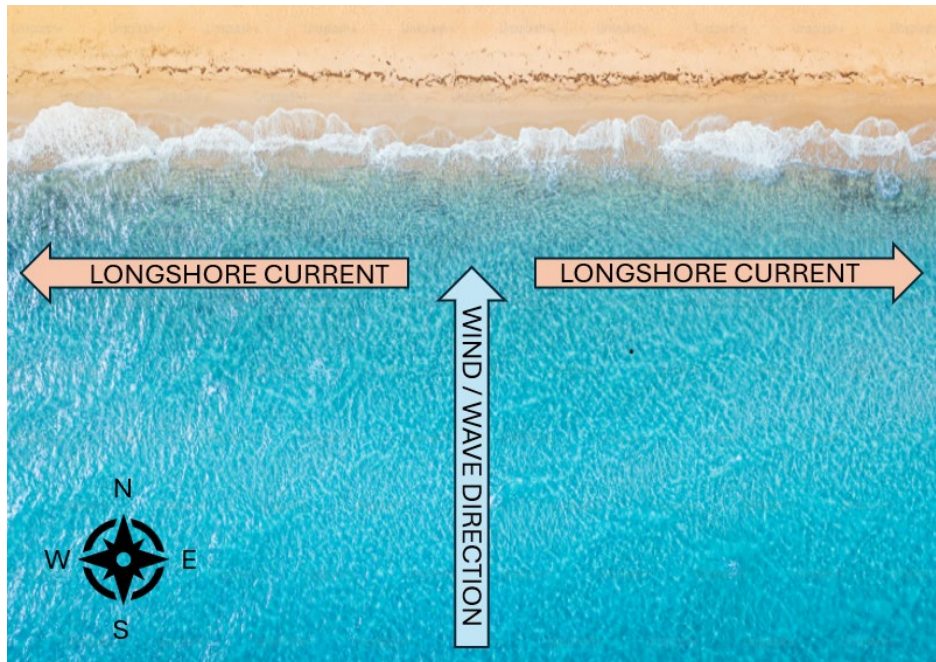


Surging: Surging breakers are the least common type of breaker. The wave builds like a plunging breaker, but the very steep-beach prevents breaking offshore. The steepness of the beach causes the building breaker to surge onto shore with significantly more energy than a spilling breaker.

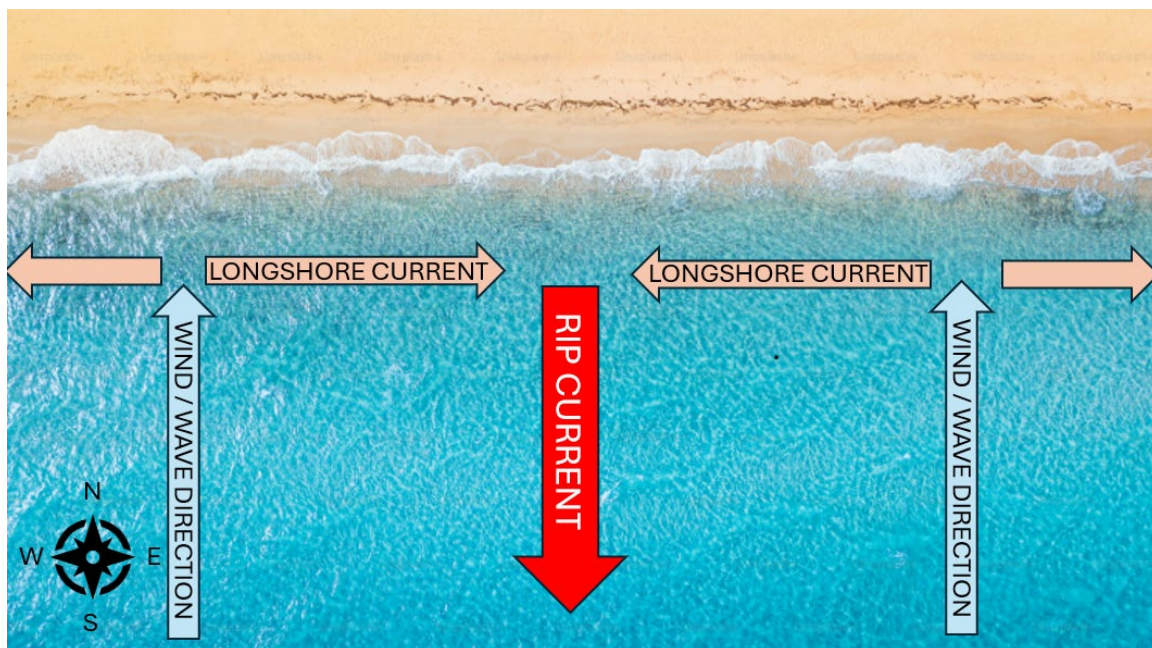


Longshore current: The longshore current is a current that runs parallel to the coast that can force landing craft to broach, an involuntary change in a vessel's course. Imagine an East-West running beach and Waves coming from the Southeast moving Northwest. The Longshore current would be moving West parallelling the beach as the waves energy is forced to continue in that direction.





Rip currents: Rip currents flow out from shore in narrow “rips” and are found on almost every open coast. They are caused by the longshore currents converging near the shoreline and creating a new current that is flowing away from shore toward open water. Once rip currents have formed, they remain constant in position until the wave conditions change. Rip currents can be seen on imagery as a calm area between breakers.



Exercise / Practice: Littoral Zones / Breakers

Sea State

Sea state is a scale categorizing sea conditions based on wave height for the free surface on a large body of water with respect to wind and swell waves. The Beaufort Wind-Force Scale is the standard for determining wind wave heights based on wind speeds ranging from 0-12. For areas outside the fetch, waves have minimal influence from the surface winds. This requires an observer to determine its Sea State based on a separate Sea State Code. Different parts of the world use different standards to measure Sea State. The most common is the Bowditch Wave Height Scale. The Department of Defense uses Bowditch Wave Heights scale to its corresponding State Code ranging from 0 to 9. Higher Sea State values have a direct correlation to operational impacts. The higher the Sea State, the more operational impacts.

Beaufort Wind-force Scale/Number	Wind Velocity		Wind Description	Descriptions of Beaufort Wind-force-generated Waves	Sea-state Code
	Knots	Miles per Hour			
0	<1	<1	Calm	Sea surface smooth and mirror-like	0
1	1–3	1–3	Light Air	Scaly ripples, no foam crests	1
2	4–6	4–7	Light Breeze	Small wavelets, crests glassy, no breaking	2
3	7–10	8–12	Gentle Breeze	Large wavelets, crests begin to break, scattered white caps	3
4	11–16	13–18	Moderate Breeze	Small waves, becoming longer, numerous white caps	4
5	17–21	19–24	Fresh Breeze	Moderate waves, take longer to form, many white caps, some spray	5
6	22–27	25–31	Strong Breeze	Larger waves, white foam streaks off breakers	6
7	28–33	32–38	Near Gale	Sea heaps up, white foam streaks of breakers	7
8	34–40	39–46	Gale	Moderately high, waves of greater length, edges of crests begin to break into spindrift, foam blown streaks	8
9	41–47	47–54	Strong Gale	High waves, sea begins to roll, dense streaks of foam, spray may reduce visibility	9
10	48–55	55–63	Storm	Very high waves, with overhanging crests, sea white with densely blown foam, heavy rolling, lowered visibility	
11	59–63	64–72	Violent Storm	Exceptionally high waves, foam patches cover sea, visibility more reduced	
12	64 and over	73 and over	Hurricane	Air filled with foam, sea completely white with driving spray, visibility greatly reduced	

Accurate oceanic and coastal analysis is crucial for maritime operations and navigation. Mariners rely on ship routing and meteorological data to avoid areas prone to severe weather. Sea state is a fundamental factor influencing maritime operations, affecting safety, efficiency, and success across various activities, from transportation and resource extraction to naval missions and rescue operations.

Cold temperatures decrease survivability of personnel in the water Survival in seawater temperatures. Sea surface temperatures more than 70°F depend more on fatigue factors than

hypothermia. Regardless, Sea surface temperatures are important for any operation of water and accurate depiction of water temperatures is required for the survivability of personnel.

Vessel Performance and Safety



Speed and Maneuverability

Rough seas increase resistance, requiring more power to maintain speed and potentially reducing maneuverability.

Stability and Navigation

Severe weather, including high winds and waves, can affect vessel stability, leading to risks like listing or capsizing. Decreased visibility and difficulty maintaining course make navigation challenging.

Crew Safety and Wellbeing

Rough seas can make it difficult for crew members to perform their duties safely, increasing the risk of injury. Furthermore, high sea states can lead to motion sickness and stress.

Operational Impacts

Submarine Missions

High sea states can reduce visibility of submarine periscopes and increase noise levels, potentially affecting sonar performance.



Amphibious Missions

Weather conditions, including sea state, are crucial for planning and executing amphibious landings, affecting troop landing, air support, and naval vessel operations.



Ship-to-Ship Missions

High waves and strong winds can compromise the precision needed for these operations.



Coastal Missions

Higher Sea State values indicate stronger wind waves, which directly impact the littoral zone and coastal operations.



Aircraft Carrier Flight Missions

Ship motion caused by a higher sea state restricts aircraft launch and recovery operations, particularly on smaller carriers.



Search and Rescue (SAR) Missions

High sea states complicate SAR operations by reducing visibility, making maneuvering difficult, and potentially increasing the risk of cold-water shock.



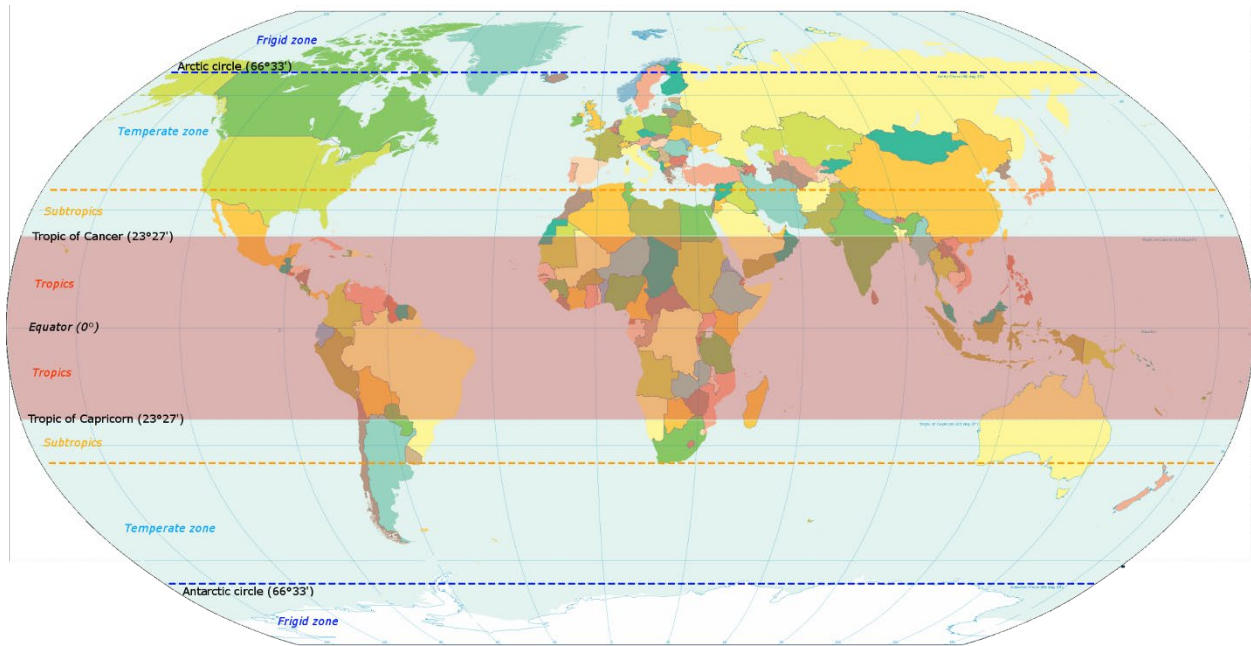
Exercise / Practice: Sea State

Introduction to Tropical Weather

General Characteristics

Meteorologically, the tropics are between the northern and southern hemisphere's subtropical highs. The tropics have a hot, humid climate with two seasons based on precipitation. The dry

season has a minimal amount of precipitation whereas the wet season has a maximum amount of precipitation. In the North Hemisphere, the dry season is our winter and spring months while the wet season is during our summer and autumn months.

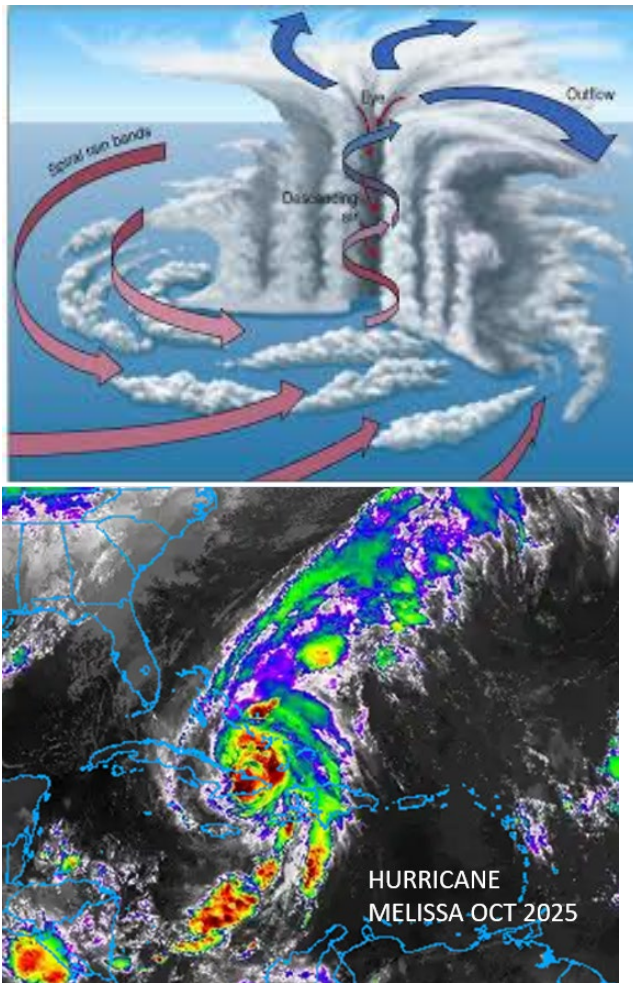


Tropical Cyclones

Tropical cyclones are non-frontal, low-pressure centers in the tropics. They are associated with rotating, organized cloud systems composed mostly of thunderstorms. They develop mostly during the wet season. In the Atlantic, Hurricane season runs from June 1 – November 30. The eastern Pacific season begins slightly earlier on May 15. The most active period within these months is typically late summer and early fall, peaking around September.

Cyclogenesis: Tropical cyclogenesis requires a large oceanic area with all the following present:

- Weak vertical wind shear less than 20 knots
- Sea surface temperature of 26°C or greater
- Warm water with a depth of 200 feet or more
- A pre-existing disturbance. In other words, any form of low-level convergence
- Must exist together poleward of 5°N or 5°S so Coriolis force can create a rotating circulation.



Cyclolysis: The requirements for a tropical cyclone at any stage to undergo cyclolysis is the opposite of its cyclogenesis. The main difference is you need all the ingredients for cyclogenesis but only need one for cyclolysis.

Stages of Development

According to the United States meteorological community, tropical cyclones have four stages of development. They are tropical disturbance, tropical depression, tropical storm, and hurricane.

Tropical Disturbance. A tropical disturbance does not necessarily have a cyclonic center. It is an organized area of thunderstorms generally 100 to 300 miles in diameter. It moves slowly in the tropics from east to west and it can maintain its identity for 24 hours or longer.

SEE ATTACHMENT FOR SLIDE ANIMATION

Tropical depression. Tropical depressions have a poorly defined cyclonic circulation with maximum sustained wind speeds of 33 knots or less. The forecast center responsible assigns a number to all tropical depressions.

SEE ATTACHMENT FOR SLIDE ANIMATION

Tropical Storm. A tropical storm has a well-defined cyclonic circulation with maximum sustained surface winds of 34 to 63 knots. When a tropical depression becomes a tropical storm, it gets a name. For example, Tropical Storm Michael.

SEE ATTACHMENT FOR SLIDE ANIMATION

Hurricane. Hurricanes have a well-developed cyclonic circulation with multiple isobars. It may contain an eye which is a cloud free area in the center of the storm with relatively calm winds and an average diameter of 20 miles. Around the eye, hurricanes have maximum sustained wind speeds of 64 knots or greater and they maintain the same name they had as a tropical storm. These tropical cyclones have different names throughout the world.

- North America (Hurricane)
- Asia (Typhoon)
- India (Tropical Cyclone / Severe Tropical Cyclone)

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Regardless of their location we measure their strength or intensity using the Saffir-Simpson scale.

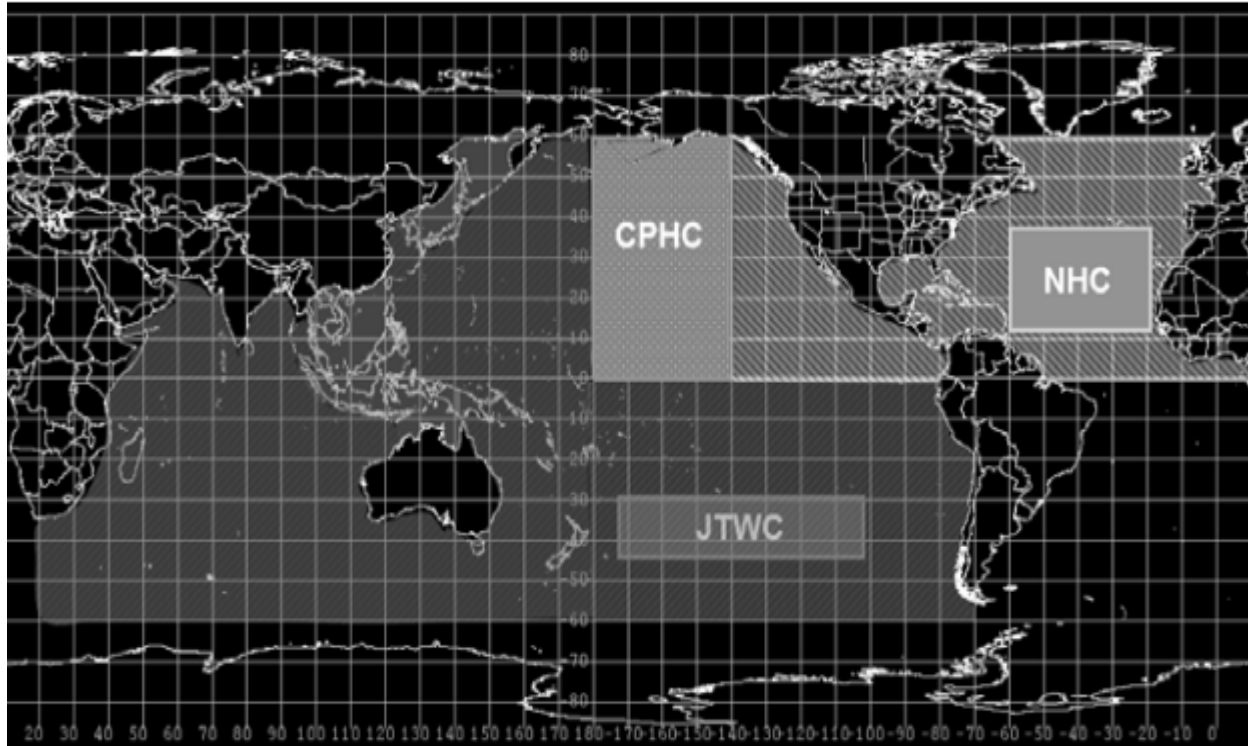
- Category 1 (64-82 KT)
- Category 2 (83-95 KT)
- Category 3 (96-112 KT)
- Category 4 (113-136 KT)
- Category 5 (137+ KT)
-

Severe Weather

While the tropical cyclone is in open ocean, the right rear quadrant contains the fastest wind speeds and heaviest rainfall. Upon landfall, the right front quadrant of a tropical cyclone contains the most severe weather. Heavy Rainfall and Storm Surge, the abnormal rise of ocean waters, are the most dangerous threat to coastal areas and is responsible for most tropical cyclone deaths. Heavy rainfall can add depth to the storm surge.

Exercise / Practice: Tropical Cyclones

Tropical Weather Agencies



National Hurricane Center (NHC) and Central Pacific Hurricane Center

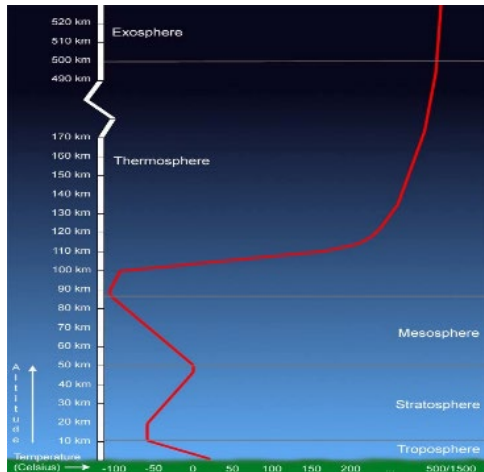
The NHC and CPHC both fall under the National Oceanic and Atmospheric Administration (NOAA). Their mission is to save lives, mitigate property loss and improve economic efficiency by issuing weather watches, warnings and forecasts for tropical storm development. They are responsible for tracking and predicting tropical weather systems. The NHC covers tropical cyclones in the Atlantic Ocean, Gulf of America, Caribbean Sea, and Eastern Pacific Ocean to 140°W. The CPHC covers the Central Pacific Basin from 140°W -180°W.

Joint Typhoon Warning Center (JTWC)

The JTWC is a joint command consisting of Navy and Air Force weather technicians. Their mission is to issue tropical cyclone warnings for DoD assets in the western North Pacific, South Pacific, and Indian Oceans. Weather technicians can modify the individual elements occurring with landfall to support their local mission but cannot deviate from the landfall timing and location forecast of the NHC or JTWC regional forecast center.

Exercise / Practice: Navigating Tropical Weather Agencies

Introduction to Space Weather **Earth's Atmospheric Layers**



Troposphere

The first and closest layer to the Earth's surface is the troposphere. The troposphere contains all weather-producing meteorological features such as clouds and precipitation. Temperatures within the troposphere cool with height meaning it is colder at the top of the layer than near the Earth's surface. The top of the troposphere is the tropopause.



Stratosphere

The second layer of our atmosphere is the stratosphere, which contains the ozone layer. It absorbs incoming ultraviolet radiation from the sun and heats the stratosphere meaning temperatures warm with height. The stratopause is the top of the stratosphere.



Mesosphere

The mesosphere is the third layer of the atmosphere. It contains the Aurora, a natural light display caused by charged solar particles entering our atmosphere near the North and South pole. The light display in the northern hemisphere is the Aurora Borealis (northern lights) and in the Southern hemisphere, it is the Aurora Australis (southern lights). Temperatures in the mesosphere cool with height, and the top of the layer is the mesopause.



Thermosphere

The fourth layer is the thermosphere, and it contains the ionosphere, which has the greatest number of electrically charged particles in the atmosphere. The ionosphere is used to reflect radio waves back to earth. Temperature warms with height due to ultraviolet radiation, and the top of the layer is the exobase.



Exosphere

The final layer is the exosphere. It gradually fades into outer space so there is no clear upper boundary for the layer. It is more important to know the temperature cools with height, and it allows atoms and molecules in our atmosphere to escape into outer space.



Space Weather Features

Insolation can significantly impact operations. The sun continuously emits electromagnetic (EM) energy outward until it interacts with the Earth's atmosphere. These EM particles between the Sun and Earth are known as the solar wind.

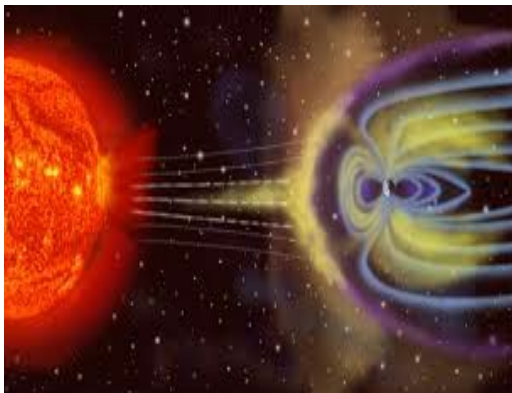
Solar Flare

A solar flare is a nuclear explosion on the sun's surface that usually occurs near a sunspot. The flare releases high-energy particles into the solar wind.

SEE ATTACHMENT FOR SLIDE ANIMATION

Geomagnetic and Ionospheric Storm

Geomagnetic storms are rapid vibrations and variations in the Earth's magnetic field are caused by the high-energy particles travelling along the solar wind. A geomagnetic storm may occur 24 to 72 hours after the solar flare and can last up to 3 days.



Space Weather Operational Impacts

High-frequency (HF) Communications

HF uses short-wave band radio waves generated by a transmitter. During transmission, the ionosphere refracts HF radio waves to a receiving ground-based antenna. The ionosphere absorbs frequencies, so they also do not refract to ground antennas resulting in a loss of communication. The amount of HF impact varies based on seasonal changes, time of day, and geographical location of the transmitting and receiving ground antennas. These change the amount of incoming solar particles. Changes also occur with increased solar activity through solar flares, increased geomagnetic activity, and variations in the solar environment.



Satellite Operations

Electrically charged particles emitted from the Sun may collide with satellites. The particles can enter, reflect, or pass through the satellite. Internal satellite systems can short circuit resulting in malfunctioning subsystems or false sensor readings. Worst case scenario is a possible loss of altitude control causing the satellite to move out of orbit or completely shut down.

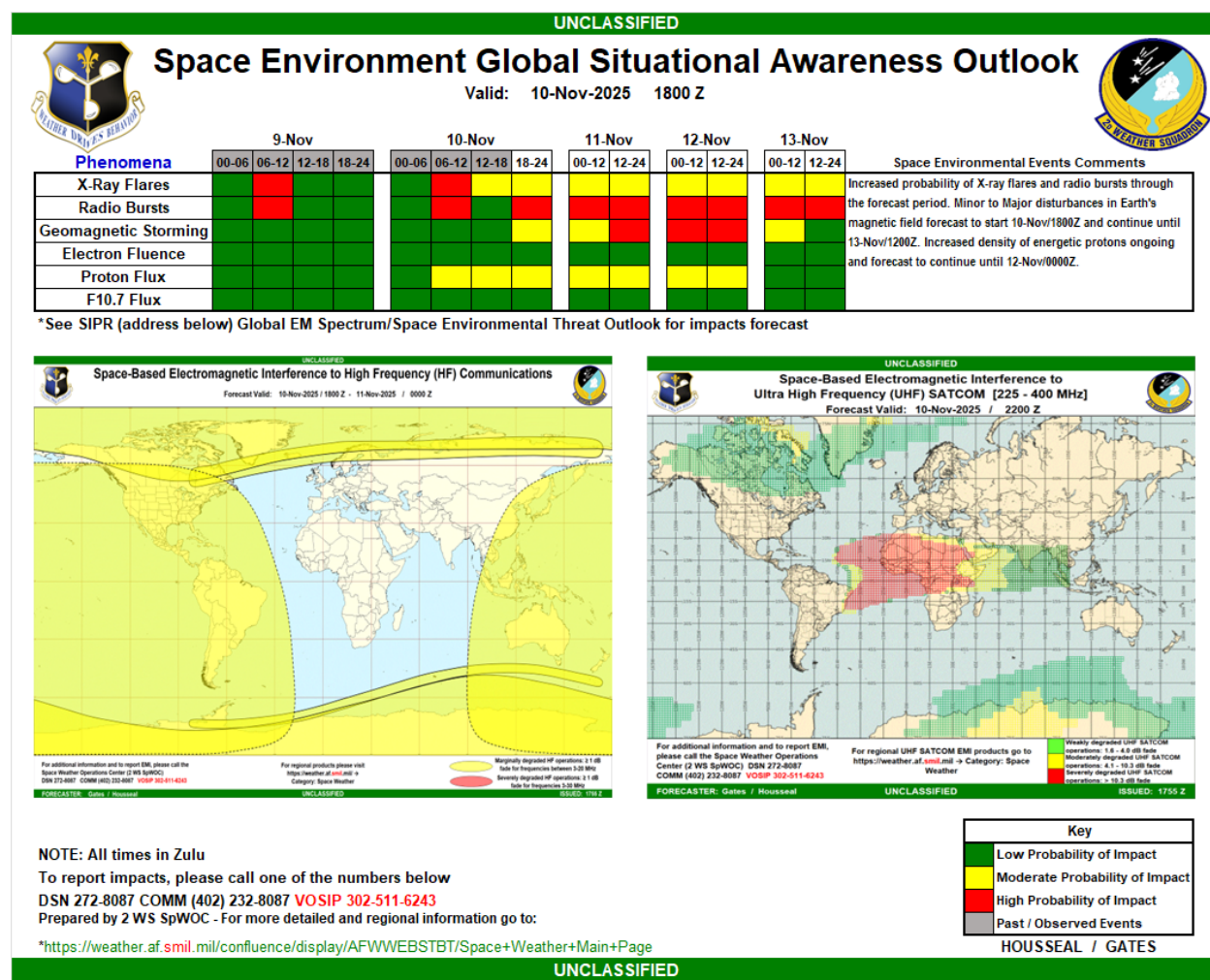


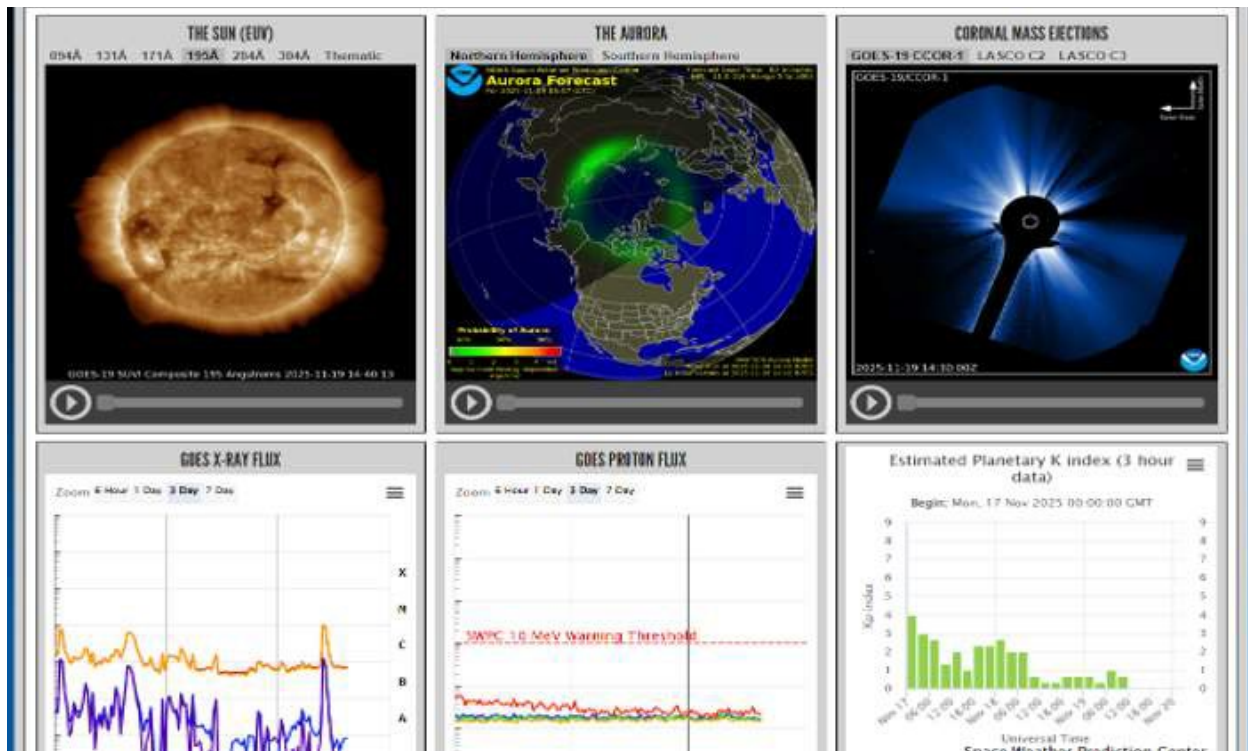
Global Positioning System (GPS)

Signals from GPS satellites travel through the ionosphere from the satellite to the receiver. The ionosphere absorbs a part of the energy which causes distortion in the signal resulting in bad or total signal loss. Potential failures include inaccurate positioning, incorrect navigation of aircraft and ships, false orientation of ground-based units, and targeting errors for GPS guided munitions.



Space Weather Products





Practice: Navigating Space Weather Impact Bulletin

PC: Ocean / Tropical / Space